

Part I

SUMMARY of Linear Approximation and Differentials

Summary

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Definition

If a function f is **differentiable** at $x = a$, then a **linear approximation to f at a** is a function given by

$$L_a(x) = f(a) + f'(a)(x - a)$$

Note 1: When the value of a is understood, the subscript is sometimes dropped from the notation. In this case, it is written as just $L(x)$ instead of $L_a(x)$.

Note 2: The graph of L is the line tangent to the graph of f at the point where $x = a$.

Note 3: If x is near a , the value $f(x)$ can be approximated by the value of $L(x)$.

Note 4: If the graph of f is concave down *on the interval* between a and x , then the approximation $f(x) \approx L(x)$ is an overestimate. If the graph of f is concave up *on the interval* between a and x , then the approximation $f(x) \approx L(x)$ is an underestimate. (The concavity has to be consistent across the interval, not just at a point.)

A **differential**, df of f at x is given by

$$df = f'(x)dx$$

Note 3: If we consider a point x and if L is the linear approximation of f at x , then for any point $x + dx$ the following holds

$$f(x + dx) \approx L(x + dx) = f(x) + f'(x)dx = f(x) + df$$

Note 4: The **increment** of f , Δf , is given by

$$\begin{aligned}\Delta f &= f(x + dx) - f(x) \approx L(x + dx) - f(x) = \\ &= f(x) + f'(x)dx - f(x) = f'(x)dx = df\end{aligned}$$

Therefore, $\Delta f \approx df$.

Part II

Recitation Questions

Problem 1

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Problem 1

- Problem 1** (a) Find the linearization, $L(x)$, of the function $f(x) = e^{2x}$ at $a = 0$.
- (b) Using the linearization, $L(x)$, from the part (a), approximate e .
- (c) Is the estimation found in part (b) an overestimate or an underestimate?
EXPLAIN.

Problem 2

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Problem 2

Problem 2 Complete steps (i)-(vii) below in order to estimate the following values using linear approximation:

(a) $\cos\left(\frac{31\pi}{180}\right)$

(b) $\sqrt[3]{8.13}$

- (i) Identify the function, $f(x)$.
- (ii) Find the nearby value where the function can be easily calculated, $x = a$.
- (iii) Find $\Delta x = dx$.
- (iv) Find the linear approximation, $L(x)$.
- (v) Compute the approximate value of the expression using the linear approximation.
- (vi) Compare the approximated value to the value given by your calculator.
- (vii) Compare dy and Δy using the value given by your calculator.

Problem 3

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Problem 3

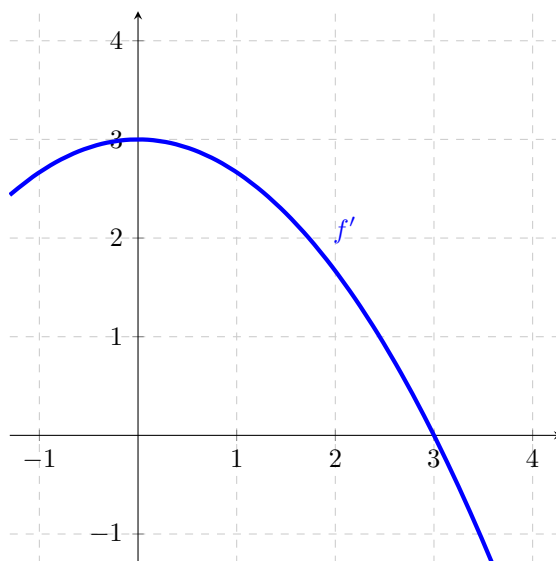
Problem 3 Estimate the value of $\sin\left(\frac{178\pi}{180}\right)$. Indicate whether your value is an overestimate or an underestimate.

Problem 4

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Problem 4

Problem 4 Consider the graph of $f'(x)$ given below. Suppose you know that $f(3) = 7$. Can you approximate $f(2.98)$ and $f(3.02)$? Explain your answer. Are these overestimates or underestimates?

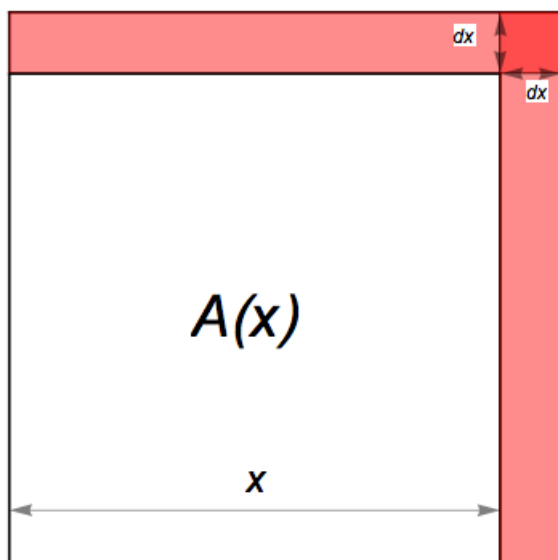


Problem 5

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Problem 5 Consider a square with a side x . Let A be the area of the square.

- Compute ΔA , the change in area if the side increases by $\Delta x = dx$.
- Compute dA , the differential of A at x , and compare it to ΔA .
- In the figure below the shaded part represents the change ΔA . Shade the part that represents dA .



Problem 6

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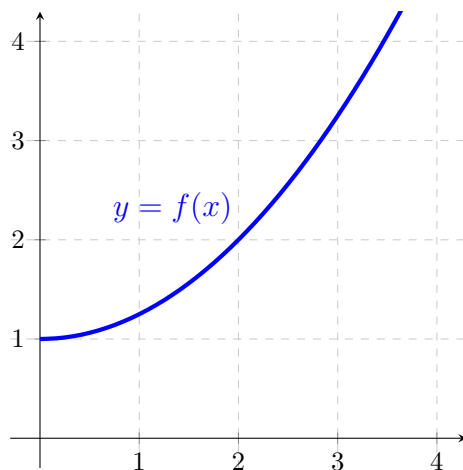
Problem 6

Problem 6 *Estimate the amount of paint needed to apply a coat of paint .05 cm thick to a hemispherical dome with diameter 50m. Is this value an underestimate or an overestimate?*

Problem 7

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Problem 7 The graph of a function f is given below.



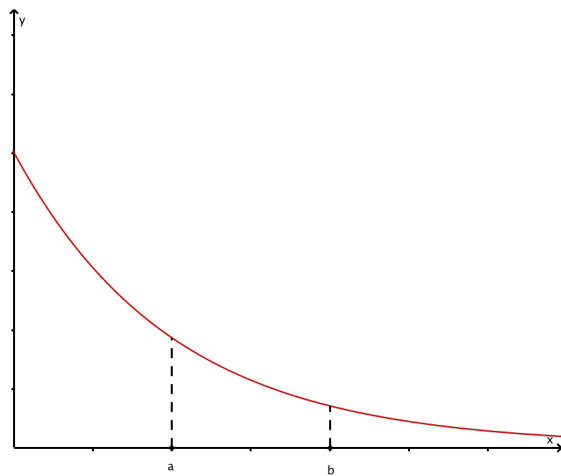
- (a) Given that $f'(2) = 1$, find the linear approximation L to the function f at $a = 2$. [2]
- (b) Sketch the graph of L in the figure above.
- (c) Use the linear approximation L to estimate the value of $f(3)$. Is this an underestimate or overestimate? **EXPLAIN**. [2]
- (d) When x changes from $a = 2$ to $a + \Delta x = 3$, the change in the **function** $y = f(x)$, Δy , is given by $\Delta y = f(a + \Delta x) - f(a)$. Draw and label Δy and Δx in the figure above.
- (e) When x changes from $a = 2$ to $a + \Delta x = 3$, the change in the **linear approximat**on, dy , is given by $dy = L(a + \Delta x) - L(a) = f'(a)\Delta x$. Draw and label $L(x)$, dx and dy (differential) in the figure above.

Problem 8

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Problem 8

Problem 8 The figure shows the graph of a function f . Let $L_a(x)$ be the linear approximation of f at a .



Circle ALL the correct statements below.

- (a) $L_a(b) < f(b)$
- (b) $L_a(b) > f(b)$
- (c) $L_a(a) < f(a)$
- (d) $L_a(a) > f(a)$
- (e) No statement (a) – (d) is correct.

Problem 9

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Problem 9

Problem 9 By using linear approximation, determine which of the following is the best estimate of $e^{0.002}$.

- (a) 1.00100050016679834166
- (b) 1.00200200133400026675
- (c) 1.00300450450337702601
- (d) 1.02020134002675581016

Problem 10

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Problem 10

Problem 10 Find a formula for the differential of the following functions.

(a) $y = 3x^6 e^x$.

(b) $z = \ln(1 + t^2)$.

(c) $\theta = \tan^{-1}(r^3)$.

Problem 11

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Problem 11

Problem 11 *In your own words, explain why $L_a(x)$ is a good approximation of the function f for x values $x \approx a$.*